

## DETAILED PROJECT PROPOSAL SUBMISSION

# DarkPulse Live Intelligence Console

A local cybersecurity intelligence dashboard for collecting, normalising, visualising, and analysing threat intelligence signals from approved sources, with repository risk scanning and clear evidence review workflows.

### MODULE CODE AND TITLE

**7COM1039-0509-2025 - Advanced Computer Science**

### DOCUMENT TYPE

**Detailed Project Proposal (DPP)**

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**Suggested file name:** 23086696-Shoaib-Latif-DPP.pdf

This proposal is prepared as a single A4 document and follows the required DPP sections shown in the assignment guidance.

## Aim of the Project

The aim of this project is to design and evaluate DarkPulse, a professional local intelligence console that helps analysts monitor cyber incidents, breach reports, defacement activity, exploit signals, social-channel intelligence, and software exposure indicators from authorised and locally controlled sources.

The system will emphasise responsible data handling, clear visualisation, reproducible scans, and safe integrations rather than unauthorised scraping or bypassing protected services.

## Research Question / Hypothesis

**Research question:** How effectively can a local integrated intelligence dashboard improve the organisation, visibility, and triage of cyber-threat signals compared with separate scripts, raw JSON files, and manual source checking?

**Hypothesis:** A unified local dashboard with structured storage, category counts, search, visual summaries, and targeted scan modules can reduce analyst effort and improve the clarity of threat triage while maintaining legal, privacy-aware data boundaries.

## Objectives

- Build a web-based command centre with consistent dashboard views for news, leaks, defacement, exploits, social intelligence, and API outputs.
- Store restored and collected intelligence records in MongoDB with clear counts, categories, timestamps, and searchable metadata.
- Implement a global impact map and live feed interface that supports quick inspection of evidence and source records.
- Provide controlled scan modules for SEO checks, GitHub repository vulnerability analysis, Play Store risk references, and authorised provider-based lookup workflows.
- Improve system usability through professional UI consistency, loading states, error messages, export options, and responsive layout.
- Evaluate the system using functional tests, response-time observations, sample scan results, and qualitative usability review.

## Short Description of the Idea

DarkPulse will operate as a local cybersecurity SaaS-style console. It will combine backend collectors, MongoDB persistence, and a dark-themed frontend dashboard. Analysts will be able to run an intelligence refresh, inspect live feed records, filter content by category, view country-level impact, and use specialised tools such as repository scanning and app-risk discovery.

The project will not depend on bypassing Cloudflare, CAPTCHA, or protected third-party pages. Where sensitive lookup services are involved, the system will use only authorised provider APIs configured through environment variables. If no authorised provider is configured, the UI will show a clear configuration message instead of presenting fake results.

## Research Methodology

### 1. Requirements and Background Reading

Review threat intelligence sharing, vulnerability scanning, dashboard usability, and ethical data-handling guidance.

### 2. System Design

Define the architecture: frontend console, FastAPI backend, MongoDB storage, collectors, scanner modules, and environment-based configuration.

### 3. Implementation

Develop collectors, API routes, scan modules, result rendering, export features, loading states, and error handling.

### 4. Testing and Evaluation

Test common workflows such as feed display, repository scan, Play Store scan, SEO report, provider configuration, and UI responsiveness.

## Considerations

- **Is data or code difficult to access?** Some sources may be protected, unstable, or legally restricted. The project will use authorised APIs, public feeds, or local datasets only.
- **Is it realistic in terms of time?** The core system is realistic if the scope focuses on working dashboard flows, MongoDB persistence, and selected scanner modules.
- **What will I learn and contribute?** The work will demonstrate full-stack engineering, threat intelligence processing, secure configuration, UI design, and responsible data workflows.
- **What is the research question?** The project investigates whether an integrated dashboard improves cyber-threat triage and analyst workflow clarity.

## Structuring and Flexibility

The proposal outlines the planned project, but implementation details may evolve after supervisor feedback. Some early answers may remain provisional because data availability, source reliability, and evaluation depth can change during development. Major changes will be discussed with the supervisor before the final project direction is locked.

## Citations

1. National Institute of Standards and Technology. 2016. *Guide to Cyber Threat Information Sharing*, Special Publication 800-150.
2. MITRE. *MITRE ATT&CK Enterprise Matrix*. Used as a reference model for describing adversary tactics and techniques.
3. OWASP Foundation. 2021. *OWASP Top 10 Web Application Security Risks*. Used to inform security-risk categories and web-facing recommendations.
4. FIRST. 2022. *Traffic Light Protocol (TLP) Version 2.0*. Used to guide responsible sharing and handling of sensitive threat information.

## Assignment Guidance Covered

### Preparation Guidance

- Start with extensive reading on the topic to inform the later literature review and research plan.
- Clearly identify the subject area and intended project work; be as specific and detailed as possible.
- Focus on answering key questions about the project.
- Avoid broad overviews of the subject area or listing every research article.
- Clearly explain the project topic, objective, and investigation method.
- Reference only a few relevant articles and keep the focus on the project.

### Submission Requirements

- Name the file using the student registration number followed by the student name and DPP, for example: **12003456-John-Smith DPP.docx**.
- Acceptable formats include Microsoft Word **.doc**, **.docx**, **.rtf**, OpenOffice Text **.odt**, **.rtf**, and Google Docs exported as **.rtf**.
- Submit the proposal as a single document file. Valid file types shown in the guidance include **doc**, **docx**, **txt**, **odt**, and **rtf**.
- Keep the DPP to approximately **2-3 A4 pages**.
- Include project title, module code and title, student name and ID, aim, research question or hypothesis, objectives, idea description, methodology, and citations.

Further guidance should be checked in the module handbook via Units > Module Information > MSc Project Handbook, and in Guidance Units > Guidance Material > DPP Guidance. It is acceptable for some early answers to be vague, but major changes should be discussed with the supervisor.